

ES647: Municipal Solid and Biomedical Waste Management

Term Paper Part II

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According to Census 2011, Aligarh city has a population of 8,74,408. Based on past year data, the average decadal growth rate for Aligarh city can be taken to be about 35.55%. Assume a design period for solid waste management as 30 years, we can estimate the waste generation. Assuming Geometric Growth Rate,

Year	Population	Growth Rate %	Average Solid Waste Generated (kg/capita/day)
2011	8,74,408	35.55	0.45 0.29
2021	11,85,260		0.52 0.34
2031	16,06,619		0.65 0.4
2041	21,77,773		0.45
2051	29,51,972		0.52
2055	33,33,895		0.56

$$P_{2021} = 8,74,408 \times \left(1 + \frac{35.55}{100}\right) = 11,85,260$$

$$P_{2055} = 8,74,408 \times (1.3555)^{4.4} = 33,33,895$$

Total Waste Generated during Design Period

Decade	Avg. Population	Solid Waste generated in Tonnes/decade
2025-35	31,530 15,76,545	21,29,124
2035-45	21,37,007	33,15,032
2045-55	28,96,714	53,39,368
		Total = 1,07,83,524

$$SW_{25-35} = 15,76,545 \times \frac{(0.34 + 0.4)}{2} = 5,03,321.65 \text{ kg/d} = 21,29,124 \text{ tonnes/decade}$$

$$SW_{35-45} = 21,37,007 \times \frac{(0.4 + 0.45)}{2} = 33,15,032 \text{ tonnes/decade}$$

$$SW_{45-55} = 28,96,714 \times \frac{(0.45 + 0.56)}{2} = 53,39,368 \text{ tonnes/decade}$$

Total Waste Generated during design period = 1,07,83,524 tonnes

$$\therefore \text{Waste Generation in 2055} = 33,33,895 \times 0.56 \\ = 1,867 \text{ tonnes/d}$$

- ⇒ Estimated waste generated of Municipal Solid Waste during the design period will be 1,07,83,524 tonnes and waste generation in 2055 will be 1,867 tonnes/d. after 30 years.
- Reason for taking geometric increase growth method for population forecasting is due to continuous migration of village people in Aligarh city and it is under development as Smart city also. So population is increasing rapidly throughout the city and that's why geometric increase method is applicable here.
- Reason for taking incremental increase for waste per capita is ^{became} ~~pack~~ use of packaged items have been increasing and as GDP increase, consumption of these kind of items producing solid waste is estimated to be increased with in a linear increments with population.
- Reason for selecting 30 years as design period is because the project for solid waste collection is handed by A2Z group for 30 years. It is also expected that technological advancements will be grow in a similar trend. so further studies would be required after design period, Further. after ~~the~~ 30 years ~~further~~ maintenance cost for machinery of handling would increase rapidly and would require investment for the same.